

TOPOLOGY-GUIDED SLIT PATTERN OPTIMIZATION TO REDUCE EDDY CURRENT DAMPING IN DIAMAGNETICALLY LEVITATED GRAPHITE RESONATORS

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Diamagnetic levitation in vacuum eliminates acoustic, clamping, and air damping entirely, making levitated resonators ideal candidates for ultra-high-Q sensing applications. Highly Oriented Pyrolytic Graphite (HOPG) is well-suited due to its large anisotropic diamagnetic susceptibility, enabling stable passive levitation above permanent magnets at room temperature. However, HOPG is an excellent electrical conductor, and as it oscillates through the inhomogeneous magnetic field of the permanent magnets, induced eddy currents damp its motion [1], limiting Q and reducing ring-down time. Alternative approaches using diamagnetic composite materials have demonstrated Q factors exceeding 450,000 by eliminating bulk conductivity [2], though at the cost of reduced diamagnetic force. Geometric modification through slit patterning preserves these properties, providing a means to engineer resonance frequencies and damping rate geometric modification in these levitated resonators.

While previous works have focused on the ability of slit geometry to reduce eddy current damping [3], here we use it to engineer the resonance frequencies of the levitated resonator. COMSOL Multiphysics finite element simulations investigate slit patterning of a HOPG plate levitated above a 2×2 NdFeB permanent magnet array. The magnetic potential energy sensitivity $|dU/dz|$ across the graphite surface reveals a characteristic 4-pointed star pattern at the plate center, identifying a low-sensitivity region where material removal minimally perturbs the vertical stiffness (Figure 1, top row). The lateral sensitivity $|dU/dx|$ reveals two high-gradient lobes along the vertical axis, with the star-shaped slit centered in a region of comparatively lower lateral sensitivity (Figure 1, bottom row). Slit patterns introduced at two threshold levels 1×10^4 and 1.9×10^4 , reduce the effective mass, increasing lateral resonance frequencies (f_{xy} and f_x) while vertical resonance frequency (f_z) remains largely unchanged (Figure 2). A topology optimization framework is additionally implemented to identify the slit geometry maximally interrupting eddy current loops while preserving levitation stiffness, with ongoing work targeting the globally optimum pattern for maximum Q improvement.

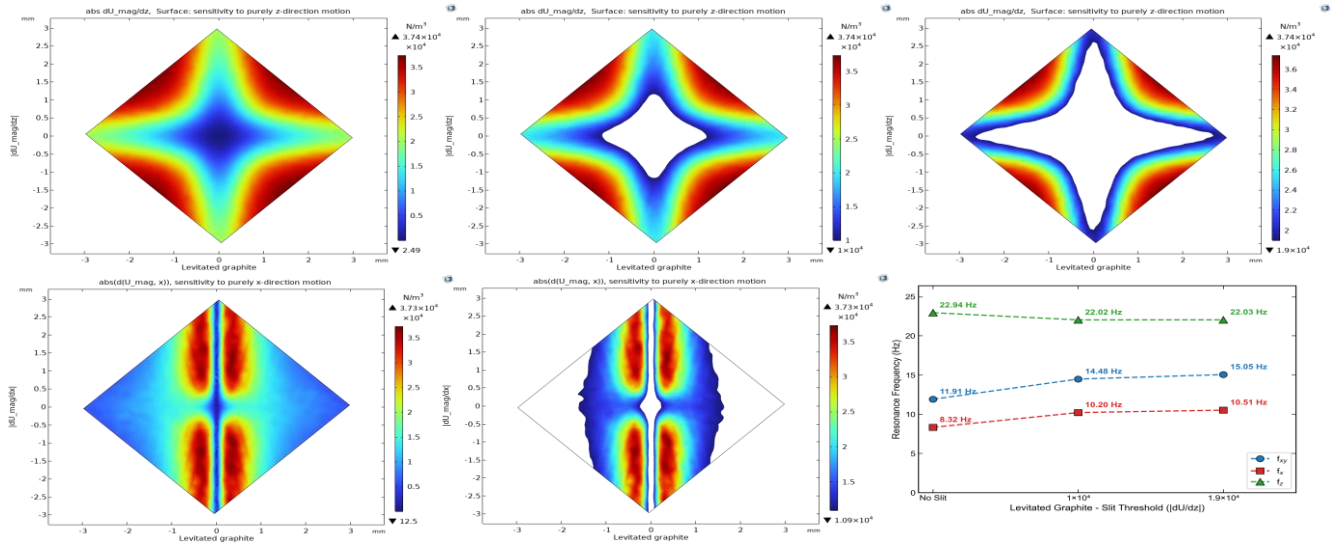


Figure 1. Normalized $|dU/dz|$ (top row) and $|dU/dx|$ (bottom row – left, center) sensitivity maps for solid graphite (left), slit at threshold 1×10^4 (center), and slit at threshold 1.9×10^4 (right) **Figure 2.** Simulated lateral (f_{xy} , f_x) and vertical (f_z) resonance frequencies as a function of threshold level.

- [1] Chen, X., Keşkekler, A., Alijani, F., and Steeneken, P.G, *Appl. Phys. Lett.* 116, 243505 (2020)
[2] Chen, X., Ammu, S.K., Masania, K., Steeneken, P.G, Alijani, F., *Adv. Sci.* 9, 2203619 (2022)
[3] Romagnoli, P., Lecamwasam, R., Tian, S., Downes, J.E., Twamley, J. *Appl. Phys. Lett.* 122, 094102 (2023).